الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري

Arab Academy for Science, Technology & Maritime Transport

Assignment: String Manipulation Programs in Assembly Language

Objective:

Develop and test assembly language programs using string manipulation and primitive instructions in the EMU8086 environment.

General Requirements

- Use string instructions: MOVSB, CMPSB, SCASB, LODSB, STOSB.
- Use SI, DI, CX registers and DS, ES segments appropriately.
- Use CLD and STD where required.
- Programs should operate on byte strings.
- Bonus: Use REP, REPE, REPNE where applicable.

Assignment 1: Block Copy (MOVSB)

Task: Write a program to copy N bytes from one array in DS to another array in ES.
Learning Goal: Understand MOVSB, SI, DI, CX, REP.
Extension: Reverse copy using STD.

Assignment 2: String Compare (CMPSB)

Task: Compare two strings and determine if they are Equal or Not Equal.
Learning Goal: Use CMPSB with REPE.
Extension: Determine which string is greater based on alphabetical order.

Assignment 3: Search Character (SCASB)

Task: Search for a character in a string. If found, return position; otherwise return -1.
Learning Goal: Use SCASB with REPNE.

Assignment 4: Count Character Occurrences
Task: Count how many times a character appears in a string.
Learning Goal: Combine SCASB with looping logic.

Assignment 5: String Length
Task: Find the length of a string ending with '$' or NULL.
Learning Goal: Use SCASB to detect the terminator.

Assignment 6: Convert to Uppercase
Task: Convert a string to uppercase.
Learning Goal: Use LODSB (read) and STOSB (write).

Assignment 7: Filter Digits Only
Task: Copy only numeric characters ('0'–'9') into a new string.
Learning Goal: Apply conditional logic with string instructions.

Assignment 8: Reverse String
Task: Reverse a string using string instructions.
Learning Goal: Use STD and pointer control.

Assignment 9: Find Substring
Task: Search for a substring inside a larger string.
Learning Goal: Combine CMPSB with an outer loop.

Assignment 10: Memory Fill (STOSB)
Task: Fill a block of memory with a specific value (e.g., 00H or FFH).
Learning Goal: Use STOSB with REP.

Note: Students are expected to test their programs and include comments explaining each step.